Bidirectional cue-constraint OT and non-equidistant sound change

- revising Boersma's BiPhon model to accommodate Korean back vowel chain shift phenomena -

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Published Date: June 17th , 2023

Abstract

With its non-teleological nature, the BiPhon model (Boersma and Hamann 2008) occupies a unique position among phonological theories on auditory dispersion. Its main operative device, namely a balancing act between articulatory and cue constraints, allows phoneme inventory to achieve an optimal state without involving any user-level intentionality. While their explanatory power is language-universal, abstract overgeneralization falls short of addressing typological variations. The model is particularly at fault in the face of non-dispersive, converging sound changes. In light of this limitation, the present work attempts to dissect the model structure and identify potential places for improvement. Locating indiscreet homogeneity in both the initial distribution of inputs and subsequent evaluation noise, relevant revisions were made. Experiments with new layouts resulted in a better capability of the model to address the non-equidistant chain shift occurring within back vowels of the Seoul Korean Dialect. Reflecting true data distribution to the evaluation noise was especially proven to be important. Predictions of the revised model are compared with real-world data samples collected over two periods.

Introduction

The preference for distinct contrast is a commonly held tendency in the history of sound change. Many languages have evolved to keep their phoneme inventory at a maximally distributed ratio along the measured auditory continuum. At the heart of this tendency lies language users' awareness of the perceptual difference between sounds, an assumption most theories on sound change share. How sound changes

utilize this awareness, however, is a matter of debate. Below introduces several examples.

Steriade's P-map correspondence theory and Flemming's distinctive constraint each tackle two competing forces in a traditional Optimality Theory framework. P-map ranks reluctance towards perceptual differences in context. Contextual neutralization arises if the violation of unfaithful mapping from the underlying to the surface form results in a greater perceptual difference in one context than the other. Meanwhile, the distinctiveness constraint incorporates perceptual awareness into the markedness constraint by rewarding constraint against a more distinct sound pair at a higher rank. Although the directionality of the usage differs, both theories assume prior linguistic knowledge in the interest of language itself outside of speakers' communicational motivation. It is impossible for a speaker to inherently know in which context perceptual difference is preferred or avoided unless the language stipulates so.

In contrast to such goal-orientedness, Wedel's exemplar theory and Ohala's theorization on misperception take the stand that auditory dispersion results from innocent misapprehension. Exemplar theory postulates that sound segments are perceived and produced through the process of storage and retrieval in our memory. During retrieval, acoustic patterns displaying the most distinguishable identity get chosen following the effort to minimize confusion. The surviving auditory signals resultingly amount to optimal contrast. There are no linguistic stipulations at play as this contrast purely stems from cognitive reasoning. Ohala's account of misperception as a mechanism of sound change is also in line with the systematically distant effort. His theory defines auditory dispersion as the result of the eventual loss of less distinct sounds (Flemming 2017). Sounds without enough distinction are subject to reanalysis as hypercorrection(overcorrection) occurs to misinterpret phonological attributes as a contextual articulatory effect (Ohala 1991).

As we should see, there was a long-standing division in academia between those that viewed the dispersive nature of sound change as a linguistically aware process or not. Boersma's bidirectional cue constraint model bridges the gap between these two conflicting stances; it starts from a place of innocent misapprehension only to ultimately optimize linguistic interest. Misapprehension here refers to not taking every auditory signal at face value, as both perception and production processes are regulated by two constraints pertinent to each domain. On the perception side, there exist cue constraints that drive input sound value to the peripheral level. In a bidirectional setting that transfers the ranking in perception to production, cue constraint alone will lead to the so-called *prototype effect*. What prohibits this extremity is articulatory constraint from the production side, which centralizes output sound values that are, in turn, used as input for the next generation. Using computer simulation, Boersma proves that repetition of this perception-production cycle throughout time will ubiquitously land sound systems with the same number of categories on the same dispersion layout. The uniformity is possible because although the theory takes linguistic consideration into interest, it avoids direct association with phonological grammar. As we shall see, cue constraint is rather closely related to a phonetic standpoint than a phonologically internal one. The departure from the strictly

phonological domain is a double-edged sword. On the bright side, the same logic can be extensively applied to all other domains of linguistics as long as bidirectional vying is valid. The downside is that it underspecifies cross-linguistic variation. Even Boersma himself only provides a minimal account of this shortfall by stating that "effects are tendencies rather than fixed rules" (223). He further tentatively claims non-optimal balance to be in a "transitory" state (223). After all, addressing variation was not much of an issue to both the phonologically linguistic P-map hypothesis and distinctiveness constraints that were context-driven.

The paper problematizes this inadequate account and attempts revision of the original model. Revisions were in light of improving its versatility to address cases that run against Boersma's hypothetical expectation. As a real-life example of such non-ideal cases, we will take a look into the back vowel shift phenomena of the Seoul Korean dialect. The back vowel shift of the Seoul dialect in recent years has witnessed a merger between /o/ and /u/vowels that rather disrupts the equidistant equilibrium of their original dispersion layout. Instead of resorting to a tentative explanation in Boersma (2008), this work acknowledges merger as the final state of movement and attributes fault to inadequate modeling. Once we revise the inadequacies, the BiPhon model can, in fact, predict such non-equidistant dispersion routes.

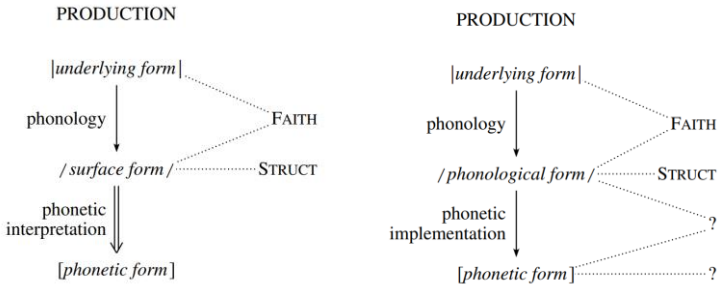
The rest of the paper runs as follows. It first probes into the operational mechanism of the BiPhon model and locates the source of its limitation. It will then proceed to the revision to solve the issue, followed by a demonstration of the revised usage. Comparing predictions of our revised model with the present sound change, I will corroborate the efficacy of my new adjustments.

BiPhon Model and its Limitations

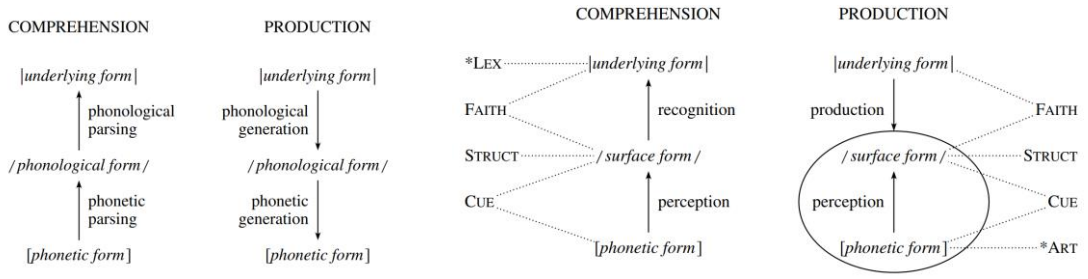
Multi-level representation framework

Before we delve into two underlying characteristics of Boersma's model; cue constraint and bi-directionality, it is important to provide some background information on where his idea stems from. Boersma's theoretic formulation would not be possible if not for a multi-level framework that spans beyond the conventionally defined domain of phonology. To this day, the mainstream account of Optimality Theory relies on a two-level framework composed of the underlying and the surface form as the figure on the left (McCarthy and Prince 1995). Given the lack of direct association of phonological surface form with physical auditory performance, this simplification was possible because 1) it delegates the duty of auditory realization to the realm of phonetics and also since 2) it assumed the universality of phonetic interpretation indicated by the double arrow compared to the single arrow between the mapping of two phonologically-pertinent levels.

It was not long before academia came to the realization that phonetic interpretation cannot be oversimplified as a language-independent universal process; hence the Hayes(1999) model involving three-level emerged (figure on the right). Nevertheless, how we would incorporate this phonetic procedure into phonology was largely left unanswered. This is indicated by the question marks in our diagram.

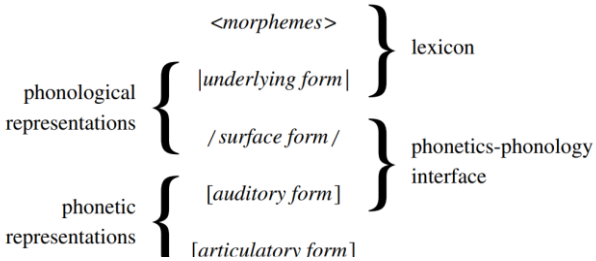


Another source of oversimplification was the model's unidirectionality. The above-mentioned OT accounts leave the perception process unattended. Mcqueen and Cutler (1997) take these two aspects into consideration in their modeling (figure on the left). As a result, we spot the incorporation of phonetic processing in both the production and perception procedures in the below diagram. This becomes the base structure of Boersma's bidirectional model (1998, 2001, 2007). The important difference, however, is that McCarthy does not implement transferability in bidirectional movements. Although both movements are defined under two analogous sequentially ordered modules, knowledge used in comprehension does not always correspond to the knowledge used in production and vice versa. In Boersma's model, this correspondence is conceptualized as what he calls a *perceptual control loop* indicated by circular confinement in the right diagram. Simply put, constraint ranking acquired during perception transfers to production. His model, thus, not only assumes bidirectionality at the observational level but also at the operational level.



His focus on this operational bidirectionality particularly lies among finer-grained levels of language. Although he ultimately posits an upward extension of the model toward the semantic interface as well, it is the model's downward extension toward phonetics that grabs the scholar's interest. As mentioned in the introduction, cue-constraint grammar gears towards a phonetic standpoint more so than a phonologically internal one. We can locate such a tendency from the fact that the BiPhon model essentially concerns the interaction between phonetic and surface forms unlike the traditional approach tackling the relation between underlying and surface forms. The former becomes the breeding ground of all research related to the

phonetics-phonology interface. Below gives a graphemic representation at full range. As I mentioned, an upward extension posits semantic-level interaction with morphemes. In order to work with variabilities at the phonetics-phonology interface, some stabilization is needed in other realms. Accordingly, the BiPhon model assumes a perfect lexicon in place that guides the gradual re-ranking of cue constraints. Cue constraint mediates the interaction of phonetics and phonology as it formalizes a pair-tied relation between acoustic signals and phonemes. Last but not least, phonetic forms can be subdivided into auditory and articulatory forms, each attending to the representation of sound and gestures. However, as with the assumption of the perfect lexicon, detailed interaction within phonetics is stabilized under universal mapping.



How does cue constraint work?

I have briefly mentioned that cue constraint is formalized as a pair-tied relation between acoustic signals and phonemes. The concept of cues displaying the connection between auditory and phonological surface form is borrowed from phonetic literature (Deguyter 59). It is a phonetic version of faithfulness mapping between phonemes and their related cues. Phonological phonemes possess standard values in their matching auditory forms such as periodicity, sonority, and formants. A vowel [a], for instance, is a periodic sonorant sound with a standard F1 value at 750 Hz. Within the same category of constraint, how closely the given constraint embodies this nature determines the degree of its ranking. Across the category, the listener weights the cue according to its discriminative capability to identify a sound segment.

In the below tableau, periodicity is weighted to a higher degree than sonority, while sonority is to F1 values. Among formants, the discrepancy between standard F1 value and phoneme increases with rankings. Since [i] and [e] each typically come with F1 of 300 and 500, connecting F1 of 800 to /i/ results in greater violation than to /e/ than to /a/. Since cue constraint probes into a paired relation, it is written in the format of *[auditory form] - /SR/.

(1) Russian perception of [a]

[*periodic*, *sonorant*, F1 = 800 Hz]	*/t/ [*periodic*]	*/b/ [*sonorant*]	*/i/ [F1=800Hz]	*/e/ [F1=800Hz]	*/a/ [F1=800Hz]
☞ /a/					*
/e/				*!	
/i/			*!		
/b/		*!			
/t/	*!				

Deguyter, p. 68

Moreover, operational bidirectionality means that the same constraint ranking learned during comprehension is re-used in production. We can thus, infer a production tableau of Russian production of [a] as below.

/a/	*[t] [periodic]	*[b] [sonorant]	*[i] [F1=800Hz]	*[e][F1=800Hz]	*[a][F1=800Hz]
[periodic, sonorant, F1= 300Hz]					*
[periodic, sonorant, F1= 500Hz]				!*	
[periodic, sonorant, F1= 750Hz]			!*		
[aperiodic, voiced (+sonorant)]		!*			
[aperiodic, voiceless (-sonorant)]	!*				

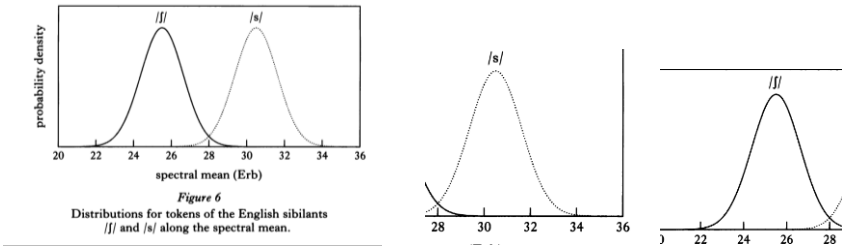
Graphical formulation of bidirectional constraint grammar

With the given understanding of how cue constraints work, the explanatory power of the BiPhon model reaches full maturity with mathematical modeling. Boersma and Hamann (2008) provide a graphical demonstration deriving evolutionary paths of sibilant inventories, which I will summarize here. The basic underlying mechanism is that mapping between auditory form and SR is guided by lexical information to maximize perceptual distance, counterbalanced by an effort to minimize articulation difficulties. Since we assume the perfect lexicon here, the model only focuses on prelexical perception as the post-lexical mapping from |UR| to <morpheme> is stabilized. In other words, in our 4-level representation (excluding the phonetic-internal domain), the model keeps the top two higher-grained relationships constant. The finer-grained relationship between sound and phoneme operates as usual.

(3) *A learner's perception tableau with reranking of cue constraints*

[26·6 Erb]	*[26·5]/s/	*[26·7]/ʃ/	*[26·6]/ʃ/	*[26·5]/ʃ/	*[26·6]/s/	*[26·7]/s/
☞ a. /s/					← *	
✓ b. /ʃ/			*!→			

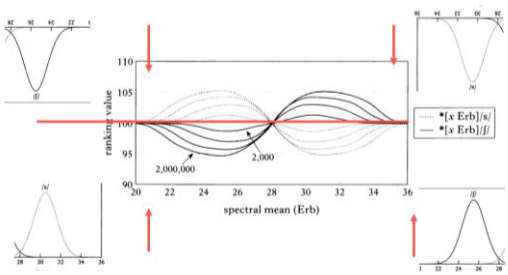
The example in the above tableau formulates cue constraints between spectral mean values and sibilants. The spectral mean value refers to the COG (center of gravity) of fricatives, which increases as the more anterior place of articulation gets. Hence, among two English fricatives /s/ and /ʃ/, the spectral mean value of /s/ is higher. We can roughly theorize the Gaussian distribution of auditory input tokens along the spectral mean spectrum consisting of two categories with mean values of 25.5 and 30.5 Erb (Equivalent Rectangular Bandwidth). The graphics of Gaussian is drawn below in Figure 6. Under this setting, auditory input of 26.6 Erb should be perceived as /s/. Therefore, in cases of erroneous identification to /ʃ/, GLA (Gradual Learning Algorithm) applies to re-rank constraint mappings between *[26.6] /s/and *[26.6] /ʃ/. What guides ranking in the correct direction is the learner's knowledge of how the surface representation should have been mapped from the lexicon.



<the cue constraint stamps>

The standard deviation of Gaussian reflects non-normalizable variations in production such as those stemming from speaker-specific characteristics or even acoustic environments of hearing. At the crux of the mathematical schematization is a reinterpretation of these distributions as inversed cue constraints. The inherent nature of cue constraint that embodies a close-knit relationship between phonetic realization and phonological SR phoneme is exploited here. As explained in the previous section, cue constraint penalizes auditory form that deviates far from the standard mapped value to the paired phoneme. The standard value here is essentially determined by its mapped frequency. A spectral mean value of 25.5 Erb, for instance, becomes the standard value of /ʃ/ as it is the most widely used cue to identify /ʃ/. The frequency distribution then, becomes inversely proportional to the rankings of cue constraints.

One can formulate this idea in graphical terms by creating cue constraint stamps with distribution graphs of each phoneme category. The stamp creates dents in initially equally ranked constraints across the input ranges. Since we are speaking of the perceptual procedure here, every spectral mean value has an equal chance to be interpreted as either category. Upon learning, the constraint stamps on each graph simultaneously affect the constraint curve concerning both sibilants. The reasoning behind this collaborative force is that acoustic cue rarely works in isolation. When an acoustic cue targets a certain frequency, the physiology of the basilar membrane stimulates neighboring hair cells that share the connection with frequencies in the vicinity of the target range. The grammar of the phonetics-phonology interface then, consists of s-shaped cue-constraint curves in the amount of category numbers.



(4) *A production tableau with cue constraints only*

/s/	*30·6 /s/	*30·7 /s/	*30·8 /s/	*31·5 /s/	*30·9 /s/	*31·4 /s/	*31·3 /s/	*31·0 /s/	*31·2 /s/	*31·1 /s/
a. [30·6 Erb]	*!									
b. [30·7 Erb]		*!								
c. [30·8 Erb]			*!							
d. [30·9 Erb]					*!					
e. [31·0 Erb]								*!		
☞ f. [31·1 Erb]										*
g. [31·2 Erb]									*!	
h. [31·3 Erb]							*!			
i. [31·4 Erb]						*!				
j. [31·5 Erb]				*!						

Following the *perceptual control loop* learning in production, this grammar transfers to production with a slight variation caused by production ranking evaluation noise. The auditory value of the lowest-ranked constraint causes the least violation, so the winning candidate of the input surface representation is the lowest point in its curve. In our demonstrative example, the chosen production spectral mean value of /s/ accordingly becomes 31.1 Erb. As we can see, the central standard value has been shifted from the original input; 30.5 Erb. The tendency of peripheral movement reflects

how the perception grammar is learned. The final width of cue constraint stamps is wider than the original distribution due to evaluation noise in perceptual ranking. Evaluation noise, by definition, refers to an erroneous term during our learning algorithm but the vague concept enters our mathematical equation as standard deviation. Intuitively speaking, the more noise there is, the less likely the learning will converge to the standard mean value. Hence, the presence of evaluation noise means that cue constraint stamps initially curved out of the input distribution, will be further dispersed. Such is reflected as increments of standard deviation in Gaussian. As a result, the location of Gaussian dents in the cue constraint curves gets modified. This is the initial effect received from the perception side.

Meanwhile, ranking evaluation noise from the production side further enhances this decentralization tendency. Similar to the perception evaluation noise, production evaluation noise also reflects as a standard deviation of Gaussian. This time, however, instead of indirectly affecting the grammar through cue constraint stamps, the noise directly affects the output distribution. This follows from the fact that erroneous term in production ambiguates the production boundary and drifts the final production values farther apart from one another. The standard deviation, therefore, increases alongside the more peripheral mean values. In the end, we can reckon the importance of evaluation noise parameters in the present computer simulation. Below shows the implementation of evaluation error logic from both sides in the Praat script code.

```
# Transmission.
#
receivedFrequency = randomGauss (producedFrequency, transmission_noise)
#
# Determine the winning candidate in perception.
#
ivalue = max (min (round ((receivedFrequency - minimum_frequency) / frequency_step) + 1, numberOfDivisions), 1)   ; determine the two viol
if decision_strategy$ = "Maximum Entropy"
        assert number_of_categories = 2
        p1 = exp (- Object_'childGrammar' [1, ivalue])
        p2 = exp (- Object_'childGrammar' [2, ivalue])
        pick = randomUniform (0, p1 + p2)
        perceivedCategory = if pick <= p1 then 1 else 2 fi
else
        if number_of_categories = 1
                perceivedCategory = imin (
                ... randomGauss (Object_'childGrammar' [1, ivalue], evaluationNoise))   ; the optimal candidate
        elsif number_of_categories = 2
                perceivedCategory = imin (
                ... randomGauss (Object_'childGrammar' [1, ivalue], evaluationNoise),
                ... randomGauss (Object_'childGrammar' [2, ivalue], evaluationNoise))   ; the optimal candidate
```

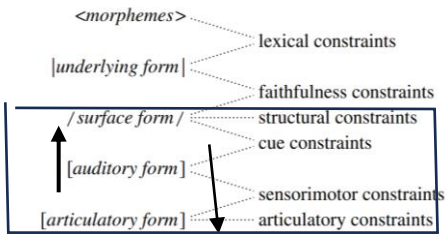
<In perception>

```
# The parent produces a phonological category. Each has equal chance.
#
intendedCategory = randomInteger (1, number_of_categories)
#
# Production noise: the parent chooses a phonetic frequency.
#
if generation = 1
        select initialLanguage
        producedRow = Draw row from distribution... p'intendedCategory'
        producedFrequency = Get value... producedRow frequency
else
        select disharmonies
        if decision_strategy$ = "Optimality Theory"
                Formula... max (
                ... randomGauss (Object_'parentGrammar' [intendedCategory, col], evaluationNoise),
                ... randomGauss (Object_'parentGrammar' [number_of_categories + 1, col], evaluationNoise))
                producedFrequency = Get time of minimum... 0 0 None
```

<In production>

At a more linguistically interpretable level, the speaker produces more extreme value than the given input because the perceptual effort attempts at mapping SR to the auditory form with the least chance to be misinterpreted as other phoneme categories. We can graphically summarize the perception–production transferring process in the sequentially connected figures of demo 1. The final production output in the rightmost figure is theorized to carry more peripheral mean values and larger standard deviations (more dispersed distribution).

Nevertheless. Cue-constraint grammar is not the sole factor influencing the production outcome. In our previously defined phonetics-phonology interface, cue constraint interacts with articulatory constraints during production. In the below-left diagram, while both cue constraints and articulatory constraints are contained within this domain, articulatory constraints intervene only during production. The extent to which each constraint is concerned with both sides of the language processing is indicated by the length of the arrows. Sensorimotor constraints are taken out of context as we leave the details of phonetic-internal interaction aside. Therefore, in production, articulatory constraints counterbalance cue constraints learned via the *perceptual control loop*. It is, in fact, intuitive to imagine physical regulation when articulating, as the extreme values obtained in demo 1 are gesturally unrealistic to produce. In contrast to cue constraints, articulatory constraints exhibit centralizing tendency. It prefers medial value at lower rank as these require minimal effort to produce. The more central rendered value gets, the less articulators like the tongue have to move beyond their neutral resting position. Since the constraint concerns auditory form solely, it shares the format *[auditory form]. In the ranking tableaux on the right, articulatory constraint *[30.8] outranks the cue constraint * [30.7] /s/. *[30.8] is the highest-ranked articulatory constraint that disrupts the cue-constraint ordering. *[30.8] thus stops the expansion of optimal value to one step below its discretized mean value in violation of the dominated * [30.7] /s/ ; 30.7 Erb.

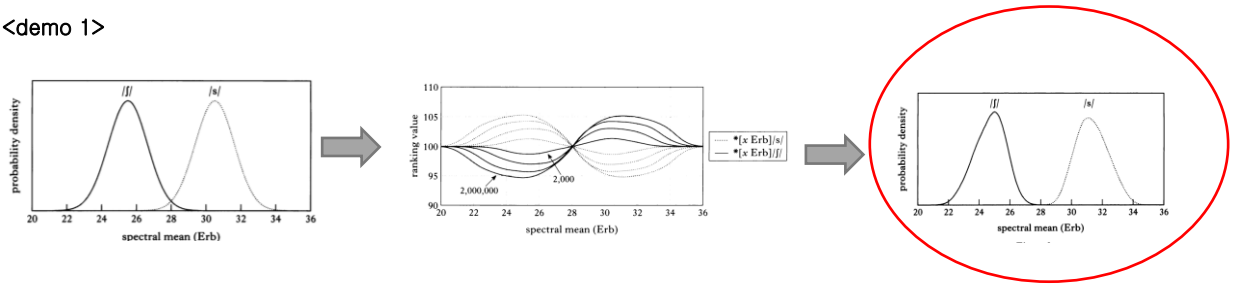


(5) *A production tableau with cue constraints and articulatory constraints*

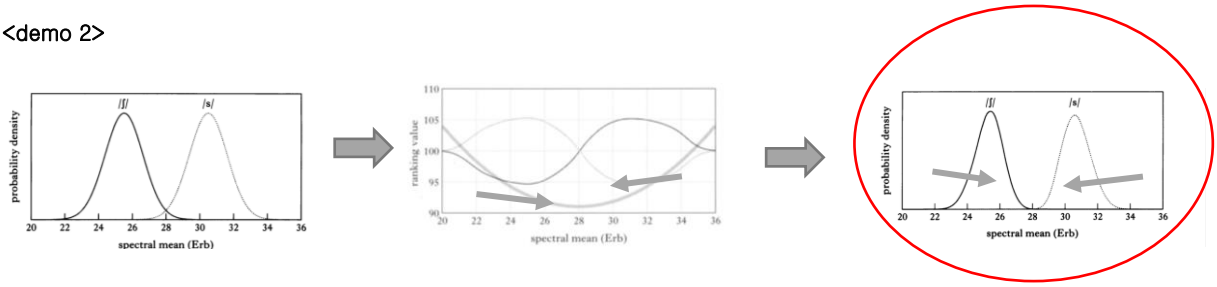
/s/	*31·2	*31·1	*31·0	*30·9	*30·6 /s/	*30·8	*30·7 /s/	*30·7	*30·8 /s/	*30·6	*30·9 /s/	*31·0 /s/	*31·2 /s/	*31·1 /s/
a. [30·6 Erb]					*!					*				
☞ b. [30·7 Erb]							*	*						
c. [30·8 Erb]						*!			*					
d. [30·9 Erb]				*!							*			
e. [31·0 Erb]			*!									*		
f. [31·1 Erb]		*!												*
g. [31·2 Erb]	*!												*	

Demo 2 gives a graphical representation of this idea. In production grammar in the middle, a bowl-shaped articulatory constraint curve is added in addition to the cue constraint curves. The molding force of the bowl curve centralizes the grammar and affects the production output in two aspects. It first moves standard mean values to the medial position, as we have analyzed in the production tableaux. In addition, it regulates the increased standard deviation caused by the evaluation production noise. Boersma and Hamann (2008) term the latter as the *entrenchment effect*. We can spot the ramifications of both effects by comparing the rightmost diagram in demo2 to that in demo1. The former involves production grammar at full range whereas the latter concerns perception grammar only.

<demo 1>



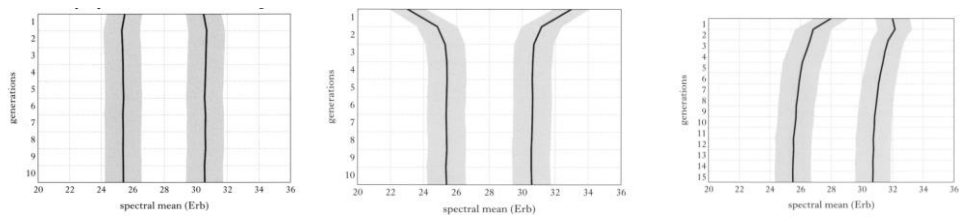
<demo 2>



So far, we have looked into the perception-to-production learning process within one generation. If we were to discuss sound change, we must view this change across generations as well. To discuss the multi-generational effect of this process, there is one additional parameter to consider. In an idealized theoretical setting, the production outcome of the former generation should be the perception learning material of the succeeding generation intact. In reality, cross-generational transmission does not take place in a vacuum. There is uncontrollable transmission noise analogous to how we initially defined production distribution i.e. between-speaker variation, background noise, etc. We must therefore reflect this noise as added distribution in Gaussian just as we did with evaluation noise.

```
"
# Transmission.
#
receivedFrequency = randomGauss (producedFrequency, transmission_noise)
"
```

The results of the simulation show that, independently of the situation in generation 1, the inventory always evolves toward the same two auditory values (249). Below is the evolutionary route of two sibilant categories with different initial set up. The graphic shows the birds-eye view of the Gaussian graphs over generations marked in the vertical axes. The thick black line notates the mean peak, whereas the gray zone area indicates the standard deviation range. The initial distribution differs in both mean and standard deviation yet uniformly lends on the same graphs at generation 10. We could infer two reasons for this. For one, the perceptual grammar induces a push chain and increases the degree of dispersion among Gaussians in proximity. On the other hand, articulatory constraint centers the distant Gaussians through a drag chain while limiting the expansion rate of dispersion. At the final state, the model resultingly arrives at the same means and standard deviations. The geometric logic thus offers a powerful explanation to address the cross-linguistic tendency in auditory dispersion. Nevertheless, the same potency that is the merit of the system also becomes the source of its limitation.



Limitation: Inexplicable variations

I have briefly noted in the introduction that Boersma's BiPhon model neglects contextual understanding in phonology. As it moves the explanatory device outside of

phonology, intricacies in the realization of surface phonological representations go overlooked. Flemming (2017) points out that the BiPhon model is unable to show neutralization or contextual enhancement. The former concerns allophonic variations, whereby a distinctive sound pair is neutralized to an alternative form in specific contexts. Take the Korean word-final stop devoicing for example. The lenis /p/-aspirated /pʰ/-fortis /p͈/ contrast is lost at the coda position as it uniformly turns into /p̚/. The loss of contrast at the syllable-final _# position can be by no means addressed in a model that only taps into the amount of sound category across all usage. How should we determine this amount if in one context the category merges while in the other it doesn't? The same question holds for contextual enhancement. If a single phoneme elsewhere has contrastive realization contextually, it is hard to simplify the number of categories in the measured dimension as one. While the lack of contextual modeling is an alarming issue, contextual amendment is inconceivable at this point, given the model's abstract, mathematical nature.

The second issue, which I will attempt revision for, is more tangible to deal with in our abstract, theoretical domain. Graphical demonstration of the previous section claims that equidistant auditory dispersion is an unavoidable fate for all sound change. Nevertheless, there are reported cases where not all categories along the measured auditory continuum become dispersed at an equal ratio. The case of Seoul Korean back vowel shift, in particular, reports a merger of /o/ and /u/ in the F1 dimension, while /ʌ/ maintains distance from the merged phoneme cluster. In the BiPhon framework, this is an inconceivable outcome, unless we tentatively assume that two vowels are in the transitory state (and will eventually move to achieve equal distance among /ʌ/-/o/-/u/). However, not only is this claim tentative and lacking evidentiality, but it also overlooks the fact that a merger can be an alternative plausible route to stabilize the sound system. Existing literature surveying sound changes in Korean such as Byun (2020) attributes the cause of the back vowel merger to the preceding front vowel merger between /e/ and /E/. /e-E/ merger primed /o-u/ merger to balance out the vowel inventory distribution. Comparing [0] and [2], one can notice how /o-u/ merger resolved the asymmetry triggered by the /e-E/ merger in [0].

(Byun 2020)

The underlying premise in Boersma's model is that optimal balance equates to maximal distribution within the delimited range. The two opposing forces in perception and production evolve to maintain an optimal distance, ungroundedly ruling out any possibility where the distance collapses and boundaries merge. Although his work initially prefaces this as a commonly observed tendency (223), the excessive potency of his model ultimately stipulates this into a fixed rule. The cautionary statement against viewing auditory dispersion as fixed rules thus becomes self-contradictory.

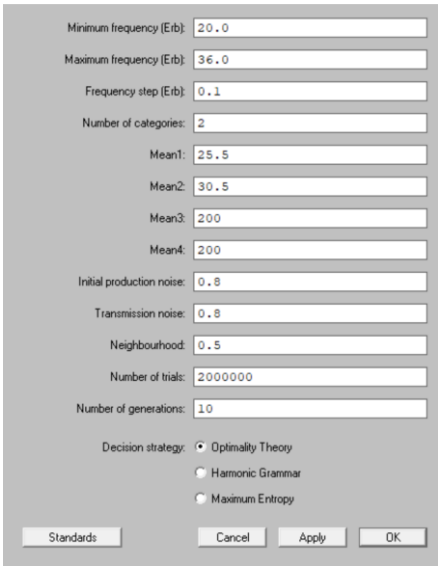
In essence, the limitation we attempt to address is the model's lack of flexibility in accounting for phoneme inventory shift that departs from Boersma's ideal

framework. The specific case we will focus on is the sound shift of Seoul Korean back vowels introduced earlier. Having identified this issue, we will now explore potential avenues for improvement

Revisions

Parameters setting

The problem at stake here is the homogeneity of cross-linguistic outcomes. The most intuitive place to resolve the lack of diversity is to change parameters. Since it was not the model's operative mechanism that was at fault, diversifying the results via diversifying input alone sounded promising. With this intention in mind, I have first identified the variables Boersma (2008) originally allowed to be changed in his model. In the below run script of his Praat modeling 1) Minimum and Maximum frequency range; 2) Number of input categories along the measured dimension; 3) Steps of discretization of continuous [auditory form] values *(Frequency step (Erb))*; 4) mean values of initial production input Gaussians *(Mean1, Mean2, Mean3, Mean4)*; 5) standard deviation of initial production input Gaussians (uniformly defined with *Initial production noise*); 6)Transmission noise; 7) Neighborhood effect coefficient (: reflects the cross-category influence that leads to the s-shaped cue constraint curves); 8)Number of training data (Number of trials; trials here means the number of randomly generated tokens that are used for training); 9) Number of generation to repeat the cross-generational learning; and finally 10) a decision strategy (: the baseline demonstration of the original paper was based on a Stochastic OT, which provides the most optimal strategy to experiment with the given framework. I have thus regarded experimenting with different strategies as irrelevant and out of topic.) were taken as variable inputs.




```
evaluationNoise = 2.0
plasticity = 0.01
```

On the other hand, there were also model parameters, where numerical input values were evidently required, yet the changes were not at the user's disposal. These

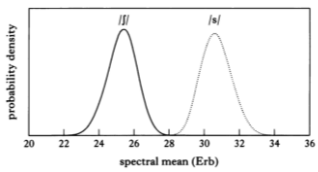
include 1) **evaluation noise of both production and perception** and 2) Plasticity (reranking steps in stochastic OT)

Scaling variables like frequency range and frequency step are not so likely to change the dynamics of the outcome as much as altering the viewing layout for convenience. Indeed, in section 7.5 of his work, the author experiments with raising the granularity of the frequency step which resulted in no change at all (262). Thus, I am excluding the importance of two out of the 11 parameters, including both those that have been *variable-lized* and those that have not. Note that the decision strategy parameter has been ruled out prior to embarking on this analysis.

The remaining 9 parameters can be classified into two groups; those that are directly concerned with the input tokens and their evolution path and those that are not. The latter group includes transmission noise, neighborhood effect coefficient, number of generations, number of training data, and plasticity. These are what I would term *global parameters*, which could affect the dynamics but only in a manner that exerts a uniform influence across categories. In other words, these are the parameters that need not be individuated because their global influence lies beyond cross-category dynamics. For this reason, excluding plasticity in the input variable set is not a significant an issue, when deriving changes cross-categorically. Note that merging can only be modeled by triggering changes between categories. We need to derive the effect of invalidating the push chain, allowing adjacent distributions to remain without necessarily moving apart from each other.

On the other hand, the former group includes the number of input categories, mean values, initial production noise, and evaluation noise. These are closely tied to the learning process of each category; hence I coined the term *local parameters*. Unlike global parameters, these **must be individuated**. In other words, an individual value is required per categorical distribution. However, there are two main issues here. The first and most apparent problem is that the model does not take evaluation as an input variable. The author fixes the evaluation noise of both production and perception with the numerical value of 2.0 across all categories. The second issue is more hidden. In the Praat operation window above, we can spot the initial production noise input slot. Probing the code, however, I discovered that initial production noise is applied uniformly across categories. This implies an abstract generalization that the real-life production evaluation noise of the initial input is identical across all categories. Truth be told, the fault was not a newfound revelation, as it was easily inferable from the writing. Boersma's simulation starts out with a Gaussian graph sharing the same width as recited on the right. In the Praat code, the identical application is instantiated by repeating the same process within the *for loop*. The remaining two local parameters, the number of categories and mean values, are justly in place.

```
# Create the initial language.
#
initialLanguage = Create Table with column names... language numberOfDivisions frequency
Formula... frequency minimum_frequency + (row - 1) * frequency_step
for category to number_of_categories
        Append column... p'category'
        factor = -0.5 / initial_production_noise ^ 2
        Formula... p'category' exp (factor * (self ["frequency"] - mean'category') ^ 2)
```



Revisiting the role of evaluation noises, they can influence the input tokens in three different stages. At initiation, initial production evaluation noise morphs the standard deviations of the generation 1 input distributions. During the perception stage, perception evaluation noise morphs the width of cue constraint stamps and influences the perception grammar, namely cue constraints. Finally, during the production stage, production evaluation noise exerts a direct influence on the output production graph by decentralizing mean values and increasing its width. Thus, it will be important to individuate input production (evaluation) noise, perception evaluation noise, and production evaluation noise (during the learning stage or from the 2nd generation). However, in the present experiment, we are focusing solely on modifying the setup of input production (evaluation) noise and perception evaluation noise.

There were three reasons for excluding production evaluation noise from the 2nd generation and beyond in this re-parameterization process. The first reason was that the production evaluation of parents was internally fixed as a global parameter in the Praat software. Below recites how production evaluation is incarnated. Upon choosing the decision strategy, the program connects to the stochastic OT algorithm, where evaluation noise is fed to the algorithm that proliferates a single fed value. Therefore, the first issue was technical difficulty. The next two reasons are more logic-based. My second concern was that changing both production and perceptual evaluation noise might be redundant and could prevent learning from converging. As I demonstrated, a Gaussian fed with evaluation noise of the perception serves as the input to production, whereby production evaluation noise is fed once more. In other words, the evaluation noise of perception and production are applied consecutively, and if the value is too large, it may exponentially complicate the process. This speculation did not arise out of nowhere. In my initial simulation attempt, I substituted Erb range with Hz, which displays a larger numerical value. The model failed to converge, prompting me to scale it down using Erb scale. I reasoned that a similar issue could occur if a significantly large individual value makes up the production evaluation noise. Thirdly, if I were to choose between perception and production evaluation noise, it would be more appropriate to prioritize perception noise that shares a direct interface with the production outcome of the previous generation. In the perception-precedes-production learning framework, perception grammar initiates the learning process. Therefore, individuating perception evaluation noise will be more effective to lay the accurate foundation of the learning system.

```
# The parent produces a phonological category. Each has equal chance.
#
intendedCategory = randomInteger (1, number_of_categories)
#
# Production noise: the parent chooses a phonetic frequency.
#
if generation = 1
        select initialLanguage
        producedRow = Draw row from distribution... p'intendedCategory'
        producedFrequency = Get value... producedRow frequency
else
        select disharmonies
        if decision_strategy$ = "Optimality Theory"
                Formula... max (
                ... randomGauss (Object_'parentGrammar' [intendedCategory, col], evaluationNoise),
                ... randomGauss (Object_'parentGrammar' [number_of_categories + 1, col], evaluationNoise))
                producedFrequency = Get time of minimum... 0 0 None
```

<production evaluation noise: unchanged>

I would also like to emphasize the appropriateness of my revision layout in that it compensates for the logical loophole that the author fails to address. In section 7.5, Boersma discusses the model's sensitivity to changes in parameters. The parameter setups he experiments with include altering the transmission noise, plasticity, number of training data, and frequency range. As the variations in these values do not amount to qualitative change, he claims the robustness of the model, leaving questions as to why any of the noise parameters were not experimented with. We could speculate that it was because the resulting outcome was different from the ideal expected route. If so, the model may lack robustness in a literal sense. He may have intentionally excluded these results because they undermine his prediction. However, if this was the case, the author has overlooked an important consideration: while non-uniform outcomes might not render the model robust against idealized scenarios, they could enhance its robustness against real-world data.

Three modification layouts

It now becomes a question of how we will implement the individuation of initial production and perception evaluation noise. As for the initial production, we can simply alter the distribution of initial Gaussians as follows. Unlike the *for loop* that uniformly applies the same standard deviations across categories, I have applied individual standard deviation values in calculating the factor part of the below code.

```
# Create the initial language.
#
initialLanguage = Create Table with column names... language numberOfDivisions frequency
Formula... frequency minimum_frequency + (row - 1) * frequency_step

#for category to number_of_categories

Append column... p1
factor = -0.5 / 2.6262 ^ 2
Formula... p1 exp (factor * (self ["frequency"] - mean1) ^ 2)   ; distribution of frequency values for the category

Append column... p2
factor = -0.5 / 1.0769 ^ 2
Formula... p2 exp (factor * (self ["frequency"] - mean2) ^ 2)   ; distribution of frequency values for the category

Append column... p3
factor = -0.5 / 1.138 ^ 2
Formula... p3 exp (factor * (self ["frequency"] - mean3) ^ 2)   ; distribution of frequency values for the category
```

Compare this with the author's original version

```
# Create the initial language.
#
initialLanguage = Create Table with column names... language numberOfDivisions frequency
Formula... frequency minimum_frequency + (row - 1) * frequency_step
for category to number_of_categories
        Append column... p'category'
        factor = -0.5 / initial_production_noise ^ 2
        Formula... p'category' exp (factor * (self ["frequency"] - mean'category') ^ 2)
```

For evaluation noise, we assume that categories with more dispersed distribution in initial input will also be more likely to maintain such tendency while learning. Exhibiting a larger standard deviation in real-world data means that learning of that category in perception is also less likely to converge to the standard value. This follows from the fact that input learning data will be largely scattered and there will be no clear guideline on how one should map each auditory form to the phonological surface representation. Therefore, the standard deviation reflecting the real-world distribution of each category can be taken as the evaluation noise value. In the below-modified code, I have once again inserted individual values used in calculating

individuated initial production Gaussian as an error term of cue constraint stamps. The given formula calculates the lowest point of the cue constraint curves. Gaussian inside the parenthesis is the stamp which the curve is molded with. Since the data I am experimenting with consists of three categories, I have left codes attending to different scenarios unattended. One can spot a comparison between blocks of *if* codes that were attended and not. In the unattended two-category scenario, for instance, globally imparted evaluation noise is consistently applied across categories. Such is an inaccurate reflection of the perception learning environment.

```
if number_of_categories = 1
        perceivedCategory = imin (
        ... randomGauss (Object_'childGrammar' [1, ivalue], evaluationNoise))   ; the optimal candidate
elsif number_of_categories = 2
        perceivedCategory = imin (
        ... randomGauss (Object_'childGrammar' [1, ivalue], evaluationNoise),
        ... randomGauss (Object_'childGrammar' [2, ivalue], evaluationNoise))   ; the optimal candidate
elsif number_of_categories = 3
        perceivedCategory = imin (
        ... randomGauss (Object_'childGrammar' [1, ivalue], 2.6261),
        ... randomGauss (Object_'childGrammar' [2, ivalue], 1.0769),
        ... randomGauss (Object_'childGrammar' [3, ivalue], 1.138))   ; the optimal candidate
elsif number_of_categories = 4
```

Application

Korean Back Vowel Shift Phenomena

Korean back vowel shift in recent years has been investigated by many scholars to occur in a merging direction between /o/ and /u/ vowel along the F1 dimension. Unlike the /e/-/E/ merger that occurred earlier, /o/-/u/ still holds a discriminative power in perceptions. The reason is that it has instead assigned the primary F1 cue to other sound measurements. Opinions are divided among those who find this along the F2 dimension and those who identify it to be a secondary cue outside of formant domains such as F0 and spectral tilt. The former view supports the argument that back vowels are undergoing a chain shift, as illustrated by [3] and [1] in the below figure. Starting with [0] as the initial state, the raising of /o/ pushes /u/ forward, creating a roughly equal distance among /ʌ/-/o/-/u/ along F2. The latter view aligns with the idea that merging itself stabilizes the vowel inventory system, eliminating the need for further movement of adjacent vowels. Amid this debate, instead of relying on data from previous studies each tailored to merit their divided opinion, I conducted a first-hand analysis over two periods and compared the prediction of the model with results derived from thereon.

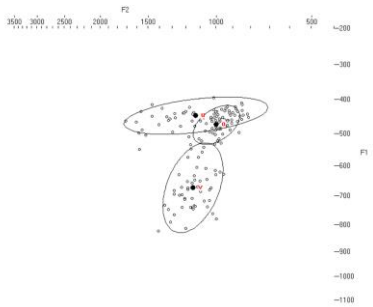
<Byun 2020>

First-hand analysis

Since the re-parameterization of our experiment can be only best tested out with details of real-world data distribution, I have conducted a first-hand analysis of formant values of back vowels over two periods. The first period is the year 2004. The corpus used here is L2KSC, which primarily features L2 Korean spoken by foreign learners. It also includes L1 Korean for comparisons, as this was the material from which back vowel formants were extracted. The population age group is young females in their 20s to 30s. I have limited the gender to females since young females were reported as the leading demographics of sound change.

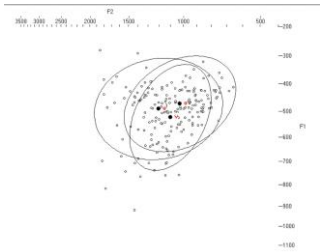
Below is the distribution tableaux of formants. The frequency in Hertz was converted to ERB (Equivalent Rectangular Bandwidth) for the sake of model convergence. The formula: ERB = 21.4 * log10(1 + (0.00437 * frequency)) was utilized in this process. The right figure plots formant values in a graph. Mean and standard deviation inside the tableaux were used as input variables.

	/ㅓ/	/ㅜ/	/ㅗ/
F1 (height) :mean, sd	674.4407 Hz, 74.90943 Hz 12.7313 ERB, 2.6261 ERB	447.3244 Hz, 28.06721 Hz 10.0256 ERB, 1.0769 ERB	474.7259 Hz, 29.64212 Hz 10.404 ERB, 1.138 ERB
F2(backness) :mean, sd	1153.112 Hz, 105.5592 Hz 16.781 ERB, 3.518 ERB	1150.643 Hz, 259.4564 Hz 16.707 ERB, 7.084 ERB	998.1651 Hz, 73.25973 Hz 15.593 ERB, 2.568 ERB



The Era 2 corpus KsponSpeech, was recorded in the year 2018. It features spontaneous speech in the Seoul dialect from speakers in their 20s and 30s. The age range and gender were fixed in line with the Era 1 corpus. The only variable between the two corpora was the time difference of recording so as to observe how real-world data distribution shifts. The vowels /ㅜ/, /ㅗ/, and /ㅓ/ were extracted manually for the acoustic measure. Given the below distribution tableaux, we can first spot a general trend of /o/ raising with a lowering of /ʌ/ along F1. In F2, /u/ moves forward, keeping three vowels somewhat equally scaled. It is also worth noting that the standard deviation greatly increases across all vowel categories. Ultimately, the experimental layout that predicts a merging trend in F1, a dispersive trend in F2, and increased standard deviation values will be the most optimal.

	/ㅓ/	/ㅜ/	/ㅗ/
F1 (height) :mean, sd	528.65 Hz, 103.96 Hz 11.08 ERB, 3.45 ERB	491.81 Hz, 99.40 Hz 10.66 ERB, 3.34 ERB	476.18 Hz, 93.16 Hz 10.43 ERB, 3 .17 ERB
F2(backness) :mean, sd	1110.52 Hz, 174.71 Hz 16.42 ERB , 5.25 ERB	1242.35 Hz, 296.96 Hz 17.28 ERB, 7.70 ERB	1040.33 Hz, 230.99 Hz 15.86 ERB, 6.49 ERB



Experimental layout

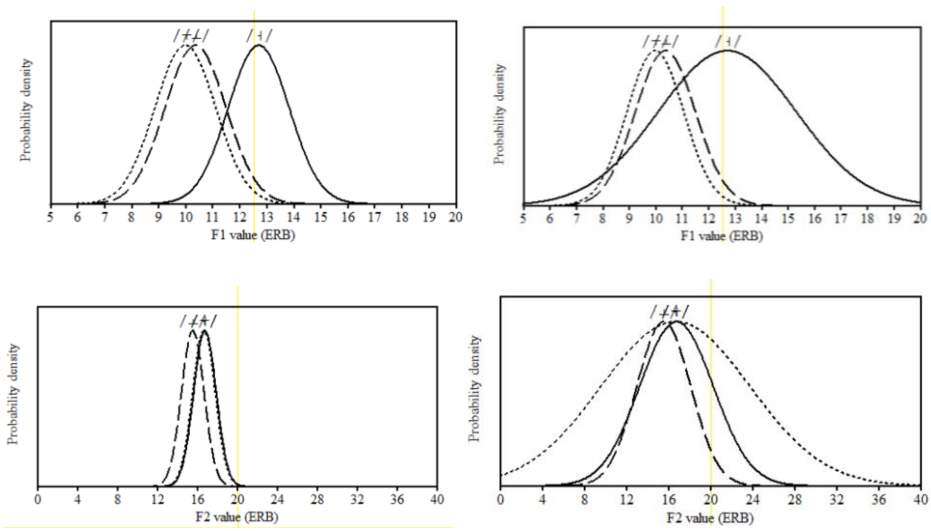
Since we are experimenting with the individuation of initial production noise as well as perception evaluation noise, the experiment is conducted following three layouts. 1)Modifying the initial production noise alone, 2) Modifying the evaluation noise during transmission 3) Modifying both. The remaining parameters mentioned in the revision section were kept at their default values.

Results and Comparison

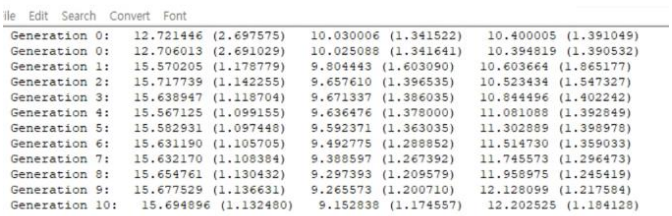
For each experiment, a bird's-eye-view graphical mapping (right) and individual Gaussian statistics for each generation (left) will be provided

1) Individuating the initial production noise alone

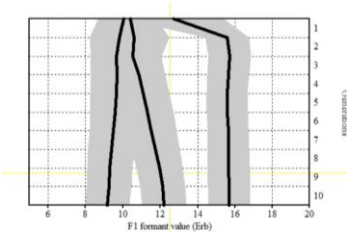
Individuating production noise will yield a change from homogenously random tokens on the left to the true Gaussian distribution on the right. One can notice the change in individual widths.



F1



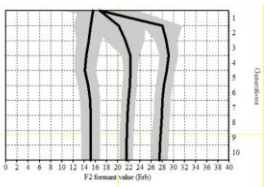
Generation 0:	12.721446	(2.697575)	10.030006	(1.341522)	10.400005	(1.391049)
Generation 0:	12.706013	(2.691029)	10.025088	(1.341641)	10.394819	(1.390532)
Generation 1:	15.570205	(1.178779)	9.804443	(1.603090)	10.603664	(1.865177)
Generation 2:	15.717739	(1.142255)	9.657610	(1.396535)	10.523434	(1.547327)
Generation 3:	15.638947	(1.118704)	9.671337	(1.386035)	10.844496	(1.402242)
Generation 4:	15.567125	(1.099155)	9.636476	(1.378000)	11.081088	(1.392849)
Generation 5:	15.582931	(1.097448)	9.592371	(1.363035)	11.302889	(1.398978)
Generation 6:	15.631190	(1.105705)	9.492775	(1.288852)	11.514730	(1.359033)
Generation 7:	15.632170	(1.108384)	9.388597	(1.267392)	11.745573	(1.296473)
Generation 8:	15.654761	(1.130432)	9.297393	(1.209579)	11.958975	(1.245419)
Generation 9:	15.677529	(1.136631)	9.265573	(1.200710)	12.128099	(1.217584)
Generation 10:	15.694896	(1.132480)	9.152838	(1.174557)	12.202525	(1.184128)



In this setup, we do not spot any significant changes along F1. It is more closely related to Boersma's hypothetical dispersion route. The only difference is in the width of the gray zone which reflects changes in initial standard deviations.

F2

```
File  Edit  Search  Convert  Font
; Generation 0:   16.781015 (3.607779)   16.863291 (6.870128)   15.593000 (2.689726)
; Generation 0:   16.803236 (3.598087)   16.869320 (6.867687)   15.580803 (2.687569)
; Generation 1:   20.197683 (4.770205)   28.018458 (3.550181)   15.183832 (3.083053)
; Generation 2:   21.392246 (3.459400)   28.778285 (2.000049)   14.866697 (2.252098)
; Generation 3:   22.341391 (1.593314)   29.207895 (1.724035)   14.418957 (2.105620)
; Generation 4:   22.329300 (1.285771)   28.791402 (1.542185)   14.158342 (1.982964)
; Generation 5:   22.186857 (1.323573)   28.315667 (1.504209)   14.771606 (1.898846)
; Generation 6:   21.950714 (1.309849)   28.073276 (1.488119)   15.069900 (1.809496)
; Generation 7:   21.823250 (1.286144)   27.946292 (1.477392)   15.169874 (1.755611)
; Generation 8:   21.658741 (1.269074)   27.754598 (1.496335)   15.195340 (1.729595)
; Generation 9:   21.606813 (1.249151)   27.625935 (1.516180)   15.207636 (1.722630)
; Generation 10:   21.567014 (1.224136)   27.499779 (1.495861)   15.156989 (1.724754)
```

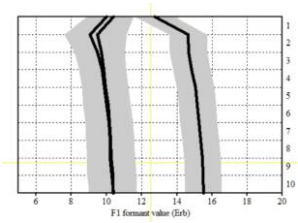


Along F2, All three setups homogenously model equal distance. The change in initial distributions is reflected in the increased standard deviation in the topmost part of the graph. The increased gray zone ratio of the /u/ vowel in particular (sd 7.048 Erb in input, slightly altered to 6.8701 after applying initial production noise) is noteworthy.

2) Individuating the perception evaluation noise alone

F1

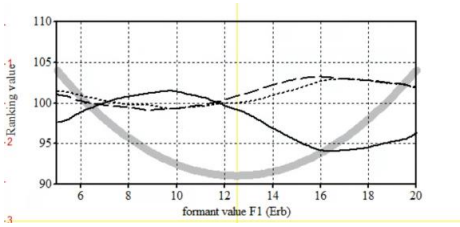
```
File  Edit  Search  Convert  Font
; Generation 0:   12.730000 (1.131371)   10.020000 (1.131371)   10.400000 (1.131371)
; Generation 0:   12.728815 (1.132512)   10.016862 (1.130455)   10.396634 (1.134700)
; Generation 1:   14.627006 (1.073364)   9.043881 (1.445260)   9.499321 (1.636419)
; Generation 2:   14.675584 (1.076112)   9.477861 (1.283223)   9.715918 (1.325145)
; Generation 3:   14.838308 (1.065431)   9.736537 (1.287099)   9.826399 (1.290270)
; Generation 4:   15.054346 (1.084419)   9.873082 (1.286012)   9.952604 (1.291637)
; Generation 5:   15.164185 (1.059235)   9.993780 (1.318954)   10.090553 (1.297271)
; Generation 6:   15.245937 (1.062534)   10.138143 (1.326423)   10.157683 (1.333756)
; Generation 7:   15.352601 (1.054355)   10.210234 (1.380449)   10.221307 (1.329861)
; Generation 8:   15.438363 (1.045909)   10.211576 (1.325137)   10.258957 (1.364441)
; Generation 9:   15.476809 (1.052533)   10.289984 (1.379430)   10.370081 (1.336355)
; Generation 10:   15.512016 (1.049271)   10.389098 (1.351788)   10.356271 (1.381160)
```

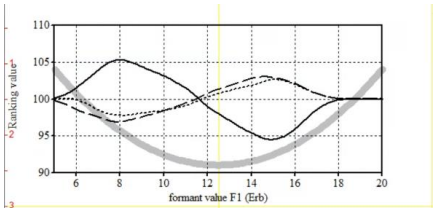


This setup approximately predicts the trend expected from our data. It predicts the merging /o/and/u/ while lowering of /ʌ/. It also models increased standard deviation for /o/ and /u/ although the increased ratio is a lot smaller. The increment ratio of standard deviation in our real-world data is summarized below for comparison.

sd	/ㅓ/	/ㅗ/	/ㅜ/
F1 -era 1	2.6261 ERB	1.138 ERB	1.0769 ERB
F1 – era 2	3.45 ERB	3.17 ERB	3.34 ERB
Increment ratio	1:1.313	1: 2.7855	1: 3.1014

To identify what is behind this altered outcome, we can take a look into the production grammar of setups 1(left) and 2(right). In setup 2, cue constraint curves in /o/ and /u/ almost overlap with one another. The way cue constraint curves are shaped is via indentation of the inversed production Gaussian graphs. If we alter the initial production noise, the effect carries on only for stamping in generation 1. Altering the evaluation noise will have more lasting effects as the altered width of the stamp creates the same dents in every cycle of perception learning. It follows then, that

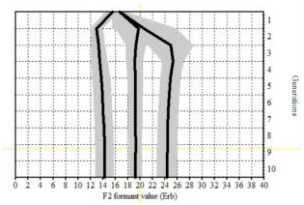




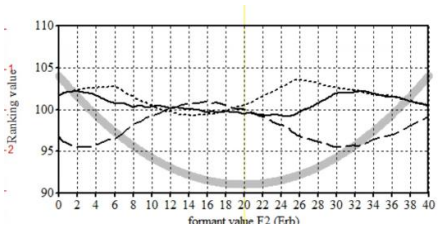
reflecting real-life distribution to evaluation noise is effective when we want to model learning environments that provide only vague guidance. After all, it is this overlap of grammar that makes initially scattered and overlapping categories converge.

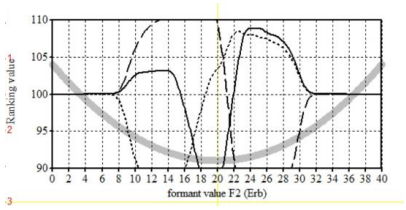
F2

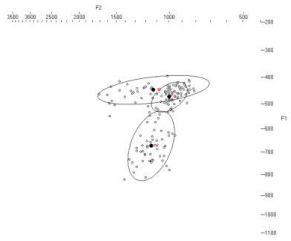
Generation 0:	16.781000 (1.131371)	16.707000 (1.131371)	15.593000 (1.131371)
Generation 0:	16.779300 (1.129074)	16.708692 (1.134818)	15.597268 (1.129700)
Generation 1:	19.762590 (3.714849)	20.680214 (5.224050)	12.978583 (1.104785)
Generation 2:	19.389919 (1.632648)	24.979010 (3.687263)	13.330307 (1.072708)
Generation 3:	19.133816 (1.326053)	25.332394 (1.733043)	13.502999 (1.334518)
Generation 4:	19.232112 (1.295085)	24.890060 (1.656849)	13.764572 (1.454061)
Generation 5:	19.287746 (1.204045)	24.709900 (1.656712)	13.929811 (1.481098)
Generation 6:	19.302763 (1.151186)	24.565555 (1.646645)	14.068569 (1.493180)
Generation 7:	19.289867 (1.131290)	24.483543 (1.659147)	14.214216 (1.498274)
Generation 8:	19.310689 (1.101533)	24.440917 (1.659500)	14.285637 (1.504644)
Generation 9:	19.288017 (1.085886)	24.397417 (1.661754)	14.337997 (1.503323)
Generation 10:	19.291063 (1.081569)	24.373975 (1.660094)	14.342029 (1.509142)

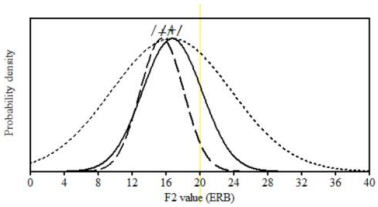


The same logic can apply to F2 perceptual grammar. Below compares the F2 grammar of setup 1 (left) with that of setup 2 (right). F2 value does not converge unlike F1 since the grammar does not overlap between /o/ and /u/. This stems from the fact that /o/ and /u/ in F1 shares a similar dispersion rate whereas in F2, the standard deviation greatly differs. I have schematized this difference below the grammar graph. Stamps from a) create dentures at the initial stage only in set up 1, whereas in set up 2, it exerts influence throughout generations consecutively. Nevertheless, the exact shape of the denture is different in every generation as the stamp used in generation n accumulates the effect up until generation n-1. We can essentially take grammar on the right as the enhanced version of the left. Besides the trajectory of the mean formant value, the increment ratio of standard deviation also moves closer to the era2 distribution. This also reflects the repeated exposure to vague learning environments.

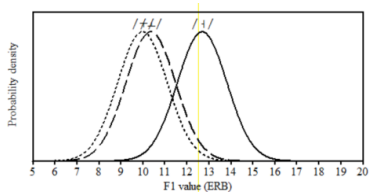








a)



3) Individuating both

In theory, we should observe the combined effect of both set up 1 and 2.

The main effect derived from setup 1 was the change in the initial width of generation 1 Gaussians. Nonetheless, the production denture only passes on to the next generation and gets diluted with the same evaluation noise as the evolution progresses. Hence, the effect of individuated distribution was not long-lived. This was attested in the Generation 10 outcome both in terms of equidistant dispersion rate and regularized standard deviation to a smaller width. The decrement ratio can be summarized as follows:

sd	/ㅓ/	/ㅗ/	/ㅜ/
F1 – generation 0	1.391049 ERB	1.341522 ERB	2.697575 ERB
F1 – generation 10	1.184128 ERB	1.174557 ERB	1.132480 ERB
Decrement ratio (F1)	1.1068 :1	1.14213 :1	2.38:1
F2 – generation 0	6.870128 ERB	3.607779 ERB	2.689726 ERB
F2 – generation 10	1.495861 ERB	1.224136 ERB	1.724754 ERB
Decrement ratio (F2)	4.592 :1	2.5136 :1	1.5559 :1

Meanwhile, set up 2 continuously applies asymmetric learning environments between categories. Larger evaluation noise means its associated category is provided with more dispersed real-life data that does not provide sufficient information to learn cue-constraint mappings. As a corollary, data with higher evaluation noise increases standard deviation to a further extent. In this vein, /u/ with the lowest F1 evaluation noise and /o/ with the lowest F2 evaluation noise each decrease the standard deviation in the according dimensions. The hypothetical increment ratio is summarized as follows:

sd	/ㅓ/	/ㅗ/	/ㅜ/
F1 – generation 0	1.131371 ERB	1.131371 ERB	1.131371 ERB
F1 – generation 10	1.381160 ERB	1.351788 ERB	1.049271 ERB
Increment ratio (F1)	1: 1.220784	1: 1.1948	1.0782 :1
F2 – generation 0	1.131371 ERB	1.131371 ERB	1.131371 ERB
F2 – generation 10	1.660094 ERB	1.081569 ERB	1.509142 ERB
Increment ratio (F2)	1: 1.4673	1.046 :1	1: 1.334

Meanwhile, categories sharing similar means and evaluation noise merges as this information is accumulatively learned in perception grammar over generations.

In summary, we are expecting 1) asymmetric gray zone width between categories at the topmost figures in the evolution graph as well as 2) convergence of /o/-/u/ mean values and overall standard deviation increments in proportional to the evaluation noise.

F1

```
; Generation 0:   12.721446 (2.697575)   10.030006 (1.341522)   10.400005 (1.391049)
; Generation 0:   12.723730 (2.707999)   10.034450 (1.337438)   10.406774 (1.391032)
; Generation 1:   15.621179 (1.158277)   9.951130 (1.664381)   10.346370 (1.796912)
; Generation 2:   15.797724 (1.133949)   10.051888 (1.424489)   10.183192 (1.414129)
; Generation 3:   15.722715 (1.084859)   10.236420 (1.396313)   10.383374 (1.370493)
; Generation 4:   15.626632 (1.061273)   10.341164 (1.408208)   10.444011 (1.394622)
; Generation 5:   15.648413 (1.046001)   10.398330 (1.377238)   10.499104 (1.420703)
; Generation 6:   15.630930 (1.043878)   10.437830 (1.422416)   10.462939 (1.380287)
; Generation 7:   15.635633 (1.044076)   10.441930 (1.402582)   10.531993 (1.383037)
; Generation 8:   15.636821 (1.051684)   10.419319 (1.401926)   10.532258 (1.399205)
; Generation 9:   15.653830 (1.038482)   10.451398 (1.375071)   10.500019 (1.433507)
; Generation 10:   15.681389 (1.036179)   10.433240 (1.393730)   10.500351 (1.388997)
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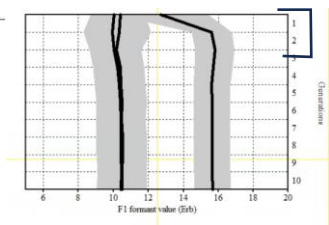

As we expected, the evolution graph here is largely similar to that of setup 2 in terms of mean values dispersion and standard deviation increments. One noteworthy difference is that the vowel /ʌ/gets lowered further down the F1 dimension with its increased value (15.512 → 15.6813). This results from the steeper push chain observed between the first and the second generation. The radical change route is caused by a larger initial standard deviation introduced with individuated initial production noise. In our real-world era 1 data, /ʌ/ shows the largest standard deviation in the F1 dimension. The less clustered distribution makes it more vulnerable to changes. The effects of set up 1 are thus proved to be more powerful in combination with altered evaluation noise. Individuating the production noise alone does not trigger much changes as the global evaluation noise cancels out this triggering over time. With the maintenance of customized stamping, however, reflecting real word data distribution to initial tokens brings some long-lasting effects. In the case of Korean back vowels, the collaborative force provides a clearer picture of the evolutionary route of /ʌ/. We can claim that the vowel will move the height of its articulation to stay distant with the will-be-then-merged /o-u/ pair.

```
; Generation 10:   15.512016 (1.049271)   10.389098 (1.351788)   10.356271 (1.381160)
```

(from set up 2)

F2

```
; Generation 0:   16.781015 (3.607779)   16.863291 (6.870128)   15.593000 (2.689726)
; Generation 0:   16.796977 (3.599931)   16.871404 (6.865755)   15.586199 (2.695997)
; Generation 1:   19.219523 (4.543633)   27.781347 (4.446254)   15.839659 (4.687692)
; Generation 2:   20.814262 (2.855769)   29.020867 (2.832159)   15.808838 (5.443288)
; Generation 3:   21.769433 (1.422796)   29.003720 (2.090409)   15.256663 (5.159996)
; Generation 4:   21.915503 (1.277140)   28.321581 (2.247277)   14.082565 (2.533364)
; Generation 5:   21.622619 (1.224481)   27.903512 (2.233657)   14.650107 (2.179246)
; Generation 6:   21.467788 (1.197836)   27.704948 (2.153366)   14.893030 (2.132696)
; Generation 7:   21.521471 (1.130559)   27.539347 (1.996999)   14.822161 (2.112626)
; Generation 8:   21.466485 (1.117015)   27.445506 (1.817415)   14.872031 (2.204700)
; Generation 9:   21.382597 (1.103540)   27.339073 (1.687900)   14.844523 (2.241249)
; Generation 10:   21.374603 (1.107792)   27.331891 (1.658662)   14.683697 (2.216635)
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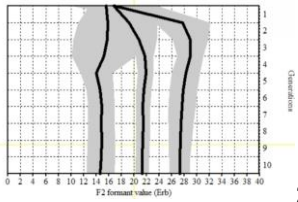

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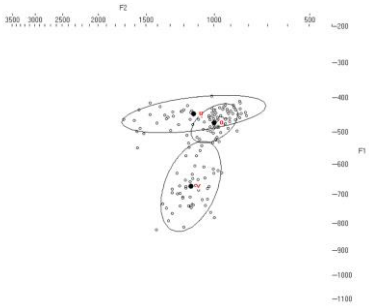
A similar evolution route is observed in the F2 dimension. The /u/ vowel with the largest standard deviation experiences a steeper push chain followed by an enhanced push chain that drives value to a more peripheral one (14.342029 → 14,683697). Other categories with evaluation noise than the default 2.0 also experience this phenomenon: /ʌ/ (24.373975 →27.331891) , /o/(19.291063 →21.374603). As a result, the between-categories distance increases while we observe how the initial production noise lives on. Among the between-categories pair, /u/ with the steepest movement drift farthest apart from its most adjacent /o/.

```
Generation 10:   19.291063 (1.081569)   24.373975 (1.660094)   14.342029 (1.509142)
```

(from set up 2)

Revisiting the formant plotting of the era 1, we can observe tendencies of sound shift. In F2, /u/ is largely dispersed, indicating a forward movement towards the anterior position of the vocal tract. In F2, /ʌ/ is largely dispersed suggesting a tongue-lowering movement upon production. These two prominent movements atr best captured in the setup 3. In the end, we conclude that it is important to individuate both the initial production noise and the perception evaluation noise. However, when testing out the effect of each alteration independently, reflecting the true distribution through initial production noise alone brings little effect without the support of individuated

evaluation noise. In contrast, reflecting the true distribution in evaluation noise alone holds enough power. It successfully predicts the general trends in sound shift, namely equidistant distribution along F2, and non-equidistant distribution along F1.



Conclusion

Upon reckoning the shortfalls of Boersma's BiPhon model, the present work revises its parameter settings and improves its capacity to accommodate non-equidistant sound shifts. The problem we attempted to solve was the model's inability to address non-dispersive mergers. The original setup uniformly landed every same category number hypothetical sound shift on the same distributions. In reality, however, phoneme inventories are a lot more diverse. This variation is fostered by the asymmetry in learning environments, where input distributions of different categories in the measured dimension differ. The asymmetry in distribution affects grammar that regulates the phonetics-phonology interface. Over time, the affected grammar accumulatively influences the sound system, a setting under which mergers can be the final state of movement. Hence, the auditory dispersion effect should not be normalized as a fixed rule.

Although our revised setting derived more lenient usage of the original model, there are still limitations to our research. The first is that our remodeling does not tackle the subject of contextual variations. To model phonological variations as such, a finer-grained revision will be required. In this endeavor, active engagement of phonology-internal knowledge will be unavoidable, which would mandate upward expansion of the BiPhon model at least up to the |UR| level. Moreover, I would like to acknowledge the incompleteness of real-world data used in this study. The distribution may not be the most accurate reflection of the status quo as they were taken from only 50 different subjects from each corpus. Also, the era 2 data was spontaneous speech as opposed to the read speech data in the era 1. This may be the cause of exponentially increased standard deviations. Nevertheless, our revised model was able to capture the general tendency of increments as well as the movement in central mean values. In the end, the main contribution to this work, using cue constraint grammar to predict non-dispersive sound shifts, stays valid.

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Software

Praat: doing phonetics by computer

with the simulation code provided on Paul Boersma's web page

*

**All non-cited sentences were written in the author's own words, reflecting her reformulated understanding of the referenced source materials.*